



OnePlus gaming triggers

OnePlus Gaming Triggers are now available to purchase in India from the official website. <http://dgit.in/may21-45>



Dell driver vulnerability

Dell has been releasing a firmware update driver since 2009 that contains "five high severity flaws". <http://dgit.in/may21-46>

are too loose or don't fit right. With all that out of the way, let's get into the different types of ANC.

FEEDFORWARD ANC

The simplest form of ANC, feedforward ANC involves a noise capturing microphone that's placed outside the headphones. This kind of ANC is most prevalent in earphones, which have limited real estate, i.e., don't have room for them on the inside. Feedforward ANC uses something called a digital signal processor to convert the noise it's hearing into anti-noise that the user will hear.

As with all kinds of ANC, there are several pros and cons to this type of ANC. With feedforward ANC, since the mic is placed externally, it has the best noise sensi-

amplify or bleed through, and the hardware can do nothing about that.

FEEDBACK ANC

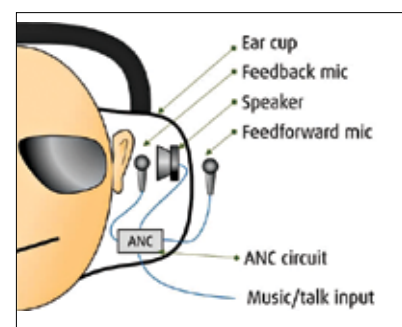
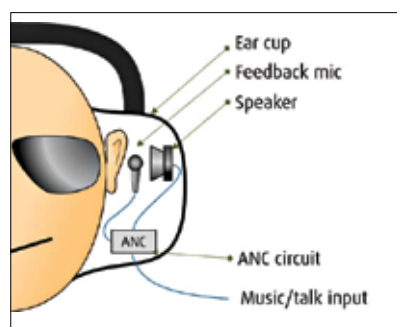
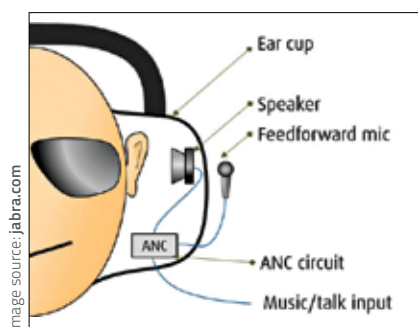
With feedback ANC, the microphone is placed inside the ear cup or earbud. While on the hardware front this presents new challenges in terms of space management, on the ANC side of things the microphone can generate more accurate anti-noise. This is because it can hear and self-correct the anti-noise it's generating. Additionally, since the microphones are on the inside, high-frequency sounds like howling winds won't affect them too much. On the flip side, its high frequency noise canceling, in general, is weaker, as the sound is less likely to get through to the microphones from the outside.

You know, that annoying shrill ringing sound you hear when you place a microphone too close to a speaker. Yes, feedback ANC has the risk of creating feedback but we're pretty certain this wasn't the reason it was named as such.

HYBRID ANC

This brings us to the third and final type of ANC, hybrid ANC. Like the name suggests, what we've got here is an ANC technology that combines both feedforward and feedback microphones and their software processing in order to get the best of both worlds.

With hybrid ANC, you've got the noise sensitivity and targeted sound isolation offered by feedforward ANC, paired with the accuracy of



tivity. This means it's the quickest to pick sounds up and therefore has more time to create an "anti-noise" signal. Its sensitivity also allows it to better tune out and isolate specific sounds, for example, it can be set up to isolate traffic or talking sounds specifically. However, the placement of the capturing mics is very important in feedforward ANC, as it's very easy for them to pick up sounds of wind, or any other sudden loud sounds, which in turn, could have the opposite effect and end up getting amplified. This is an issue with feedforward ANC as there's no way to self-correct; the device can't hear the anti-noise signal it's generating. This becomes a bigger issue when, like we mentioned above, there's problems with fitting, which could cause sounds to

Once again, a software processor is required in order to get the ANC working, this time with a new set of challenges. With the microphones being on the inside in feedback ANC, this means that the microphones can also hear the audio that you're listening to, the software needs to filter this out while also generating anti-noise and making the required corrections. With the microphones being on the inside and closer to the ears, unlike with the feedforward design, there's comparatively less time for the software to process the anti-noise.

Additionally, there's a risk that the microphones can pick up its own generated anti-noise, this would cause the noise to actually amplify, instead of the anti-noise cancelling out the background sounds. There's also a risk of this creating feedback.

feedback ANC, but without the issues that feedback ANC or feedforward ANC might suffer from. With mics on both the inside and outside, the range of frequencies it can suppress is much broader, plus it can self-correct or adapt to any issues.

The biggest con for hybrid ANC is its cost. You've got twice as many high-quality mics present in a headphone with hybrid ANC, and you need a more powerful processing software in place to handle both the feedback and feedforward aspects of hybrid ANC. The thorough testing that's required also adds to the overall cost, which ends up being much higher than headphones using the previous two types of ANC. There should be no doubt that hybrid ANC would provide the best quality of ANC though. ■